INCH-POUND
MIL-DTL-24640/25A
8 November 2011
SUPERSEDING
MIL-DTL-24640/25
9 August 2008

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 TO +105 °C, 600 VOLTS, TYPE 9XS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction, Non-Watertight

Components 1-4:

First - Component wires shall be SAE-AS81044, except print identification shall be in accordance with table II. Testing shall be in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing procedures shall be in accordance with SAE-AS81044.

Second - Two component wires twisted together. The lay length shall be in accordance with table I.

Component 5:

First - Component wires shall be SAE-AS81044, except identification shall be in accordance with table II. Testing shall be in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing procedures shall be in accordance with SAE-AS81044.

Assembly:

Third - The components 1-5 shall be cabled together with a lay in accordance with table I. Fillers may be used, as necessary, to form a round cable.

Fourth - Optional binder tape.

Fifth - Braided shield of 34 AWG tin-coated copper. Braid angle of 40 to 50 degrees; minimum coverage of 90 percent.

Sixth - Optional binder tape.

Seventh - Cross-linked polyolefin jacket with a minimum wall thickness of 0.050 inch. Color black.
NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:48Z
Check the source to verify that this is the current version before use.

TABLE I. Construction details.

Military part no. M24640/25	Type & size	Conductor size AWG	Conductor resistance	Component wire	Pair lay (inch)	Cable lay (inch)	Minimum bend radius (inch)	Overall diameter (inches)	
								Type 9XS	
								(min.)	(max.)
-01UO	9XS-12	12	2.02	SAE-AS81044/12	2.50±0.25	7.5±1.0	5	0.625	0.675
-02UO	9XS-10	10	1.26	SAE-AS81044/9	3.50±0.50	9.5±1.0	7	0.845	0.905
-03UO	9XS-8	8	0.701	SAE-AS81044/9	5.50±0.50	12.5±1.0	10	1.200	1.300

TABLE II. Color code.

Component	Insulation color	Print
1	White	Black
	White	Brown
2	White	Black
	White	Red
3	White	Black
	White	Orange
4	White	Black
	White	Green
5	White	Black

REQUIREMENTS:

Qualification required.

Temperature rating	150 °C (302 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Accelerated aging	Oven temperature, 300±2 °C (572±3.6 °F) for 6 hours
Blocking	150±2 °C (302±3.6 °F)
Flammability	30 seconds (max.)
	3 inches (max.)
	No flaming tissue paper
Humidity resistance	5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Dielectric test (100% test)	
Primary insulation	Impulse test - 6.0 kV (peak) or 3 kHz spark test - 1.5kV root mean square
Component wire	Impulse test - 8.0 kV (peak) or 3 kHz spark test - 5.5kV root mean square

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:48Z
Check the source to verify that this is the current version before use.

Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	Oven temperature 200±2 °C for 168 hours
Physical properties of primary insulation:	
Tensile strength	2500 pounds per inch (min.)
Elongation	150 percent (min.)
Shrinkage	0.125 inch (max.) at 300±2 °C (572±3.6 °F)
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch
Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	2500
Conductor to shield	2500
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.)	Not applicable
Capacitance unbalance (percent, max.)	Not applicable
Characteristic impedance (at 1 MHz, ohms)	Not applicable
Watertightness (see table IX of MIL-DTL-24640 for limits of water leakage)	Not applicable
Weight	No failure

Group B:

Cold bending, cable mandrel diameter 12 times cable diameter	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:48Z
Check the source to verify that this is the current version before use.

Drip	Not applicable
Cross-linked proof test (percent, max.)	
Jacket	50
Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	Not applicable
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged), air oven	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Heat distortion (percent of unaged, max.)	30
Permanence of printing (conductor insulation, cycles, Method 1 only, min.)	75
Permanence of printing (jacket) (cycles, min.)	125
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	Not applicable
EMP response (dB, min.)	Not applicable
Group D:	
Flame propagation (cable)	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:48Z
Check the source to verify that this is the current version before use.

Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	35
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
9XS-8, 9XS-10, 9XS-12	250

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:48Z
Check the source to verify that this is the current version before use.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-100)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:48Z
Check the source to verify that this is the current version before use.